



Welcome to the Else Kooi Laboratory

January 2023 - Cryo Lab version



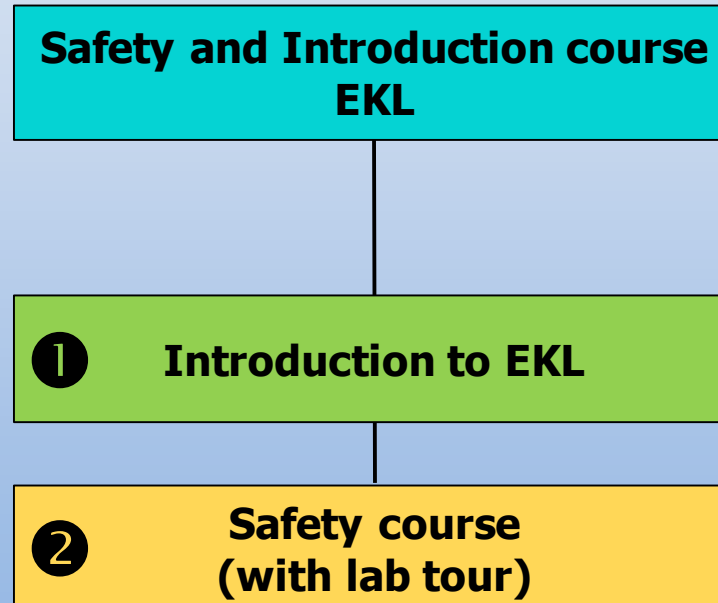
Introduction to the



EKL laboratories

(formerly known as the Dimes Technology Center)

Program for today



Outline

- 1. Introduction to EKL**
- 2. Access to the EKL labs, buddy system & start info**
- 3. Cleanroom definition & layout**
- 4. Equipment policy**
- 5. Ethics & violation of rules**

Introduction to EKL

Else Kooi Laboratory



- EKL takes its name from the Dutch chemist [Else Kooi](#) (1932-2001).
- Main objectives:
 - Facilitate excellent (sub) microfabrication.
 - Support Science & Research as well as Prototyping & Small scale production.
 - Build a bridge between academic research and industrial innovation in the microfabrication field.



Else Kooi

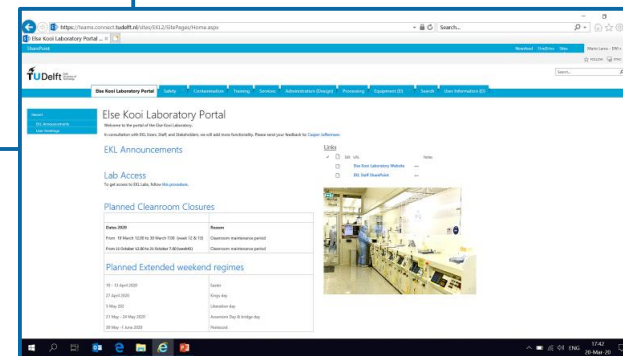
Else Kooi Laboratory

- More than **30 years** of experience.
- More than 150 researchers, PhD students, MSc students , technologists & technicians.
- Processes: Bipolar, BICMOS,
MEMS, TFT, Solar cell,
Polymer technology, Organ on a chip...
- Status of **Preferred Training Center** for many universities and companies world-wide.

SharePoint & NIS

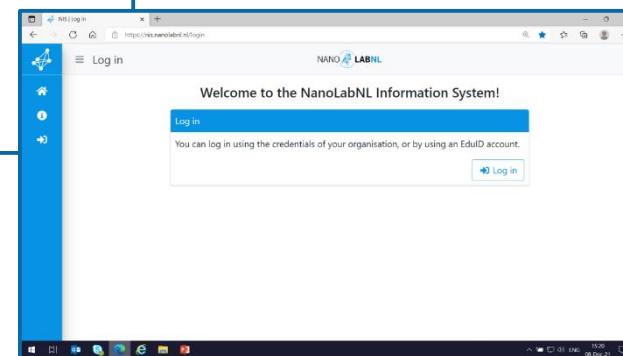
All **information** and **documents** for this training can be found on the EKL **SharePoint** website:

<https://sharepoint.tudelft.nl/sites/EKL2>



Applying for a **training** must be done through the **NanoLabNL Information System (NIS)** :

<https://nis.nanolabnl.nl>



Access to the EKL labs

The procedure for obtaining access to the EKL labs is described on the [EKL.tudelft.nl](https://ekl.tudelft.nl) website under [Lab Access](#).

Requirements:

- ➔ Attend the EKL [Safety & Introduction](#) course and pass the questionnaire.
- ➔ Acquire an account in the NanoLabNL Information System (<https://nis.nanolabnl.nl>).
- ➔ Submit a signed [EKL Access Form](#). Contact persons: [Bianca Knot](#) (secretary), [Vincent van Croonenburg](#).

When access is granted:

- ➔ Only **authorized users** may enter the EKL labs.
- ➔ Users must have a **valid badge** in order to enter/exit one of the labs.
 - It is not allowed that 2 users enter a lab with one badge.
 - It is not allowed to use the badge of another person.



Refreshment: Once per year a refreshment questionnaire must be passed to keep lab access.

This must be done in the NanoLabNL Information System. You will automatically receive a reminder.

Availability of the EKL labs

- The **MEMS lab** is open from Monday through Friday between 7 am and 20 pm.
- The EKL labs routinely shuts down two weeks per year (in April and October) for general equipment and facilities maintenance, as well as on some public holidays.
 - The lab will occasionally have intermittent closures due to a variety of reasons. These closures usually last less than a day.
 - When the facility is to be closed, advance notice will be posted on NIS as well on the Lab entrance doors.
- More specific information on when the shutdowns are planned will be communicated via the NanoLabNL information system (<https://nis.nanolabnl.nl>).

EKL Buddy System

- The Buddy system requires that a **minimum of two** people are inside the lab at all time.
- A **buddy** is another knowledgeable user within the facility, such as a staff member or a good experienced user, who can and will watch over you and care for you in case of trouble.

- Any use of the **wet benches & fume hoods** or **work with chemicals** requires the use of the **Buddy system**, no matter what time of the day it is.
- Using other equipment does not require the buddy system.



EKL Buddy System

- Each lab user is responsible to find his/her own buddy.
- Each lab user is **individually responsible** to ensure that a buddy is **ALWAYS** present during the use of wet benches and fume hoods or the work with chemicals.
- Buddies **may change** during the lab use, it does not continuously have to be the same person.



I NEED A **BUDDY**

Visitors

- Visitors can enter the laboratories **only after approval** by EKL.
- Visitors and guests must be escorted by a **qualified** staff member or **permitted** user.
- Visitors and guests are admitted to the laboratories for the purposes of a **tour** or **observation only**. Visitors will **not** be allowed to process devices or to use any of the available equipment.
- All visitors will need a **guest badge**. It is not allowed to enter the laboratories without one.

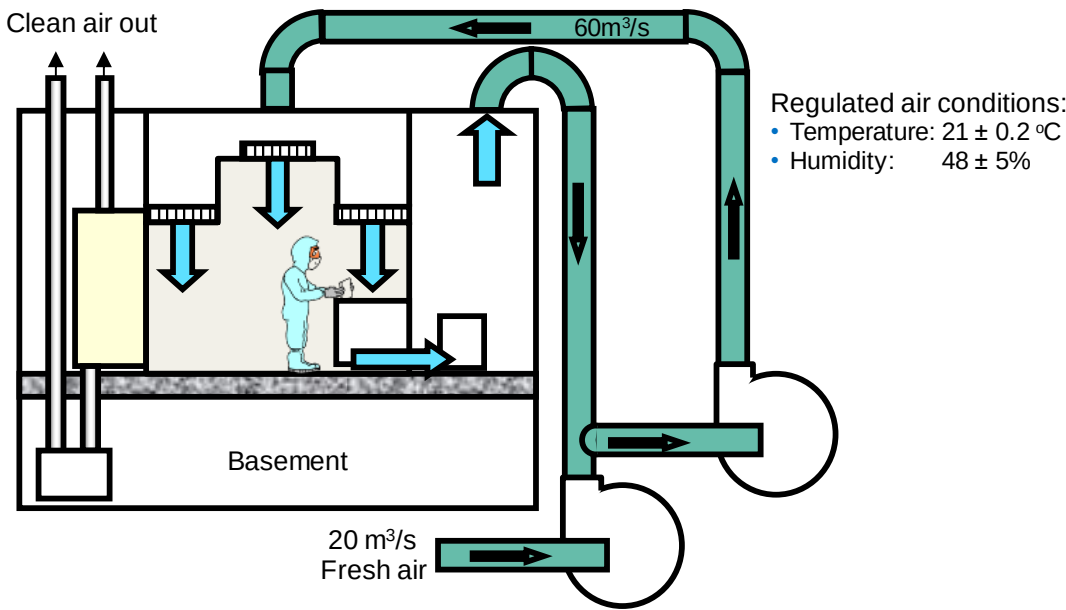
For bringing in visitors, contact: [Silvana Milosavljevic](#) or [Vincent van Croonenburg](#).

Cleanroom

A **clean room** is what your mother always wanted from you.



And a **cleanroom** is what we are discussing today.



Cleanroom definition

A cleanroom is defined as an enclosed area that is specially designed to control, monitor and maintain several parameters such as:

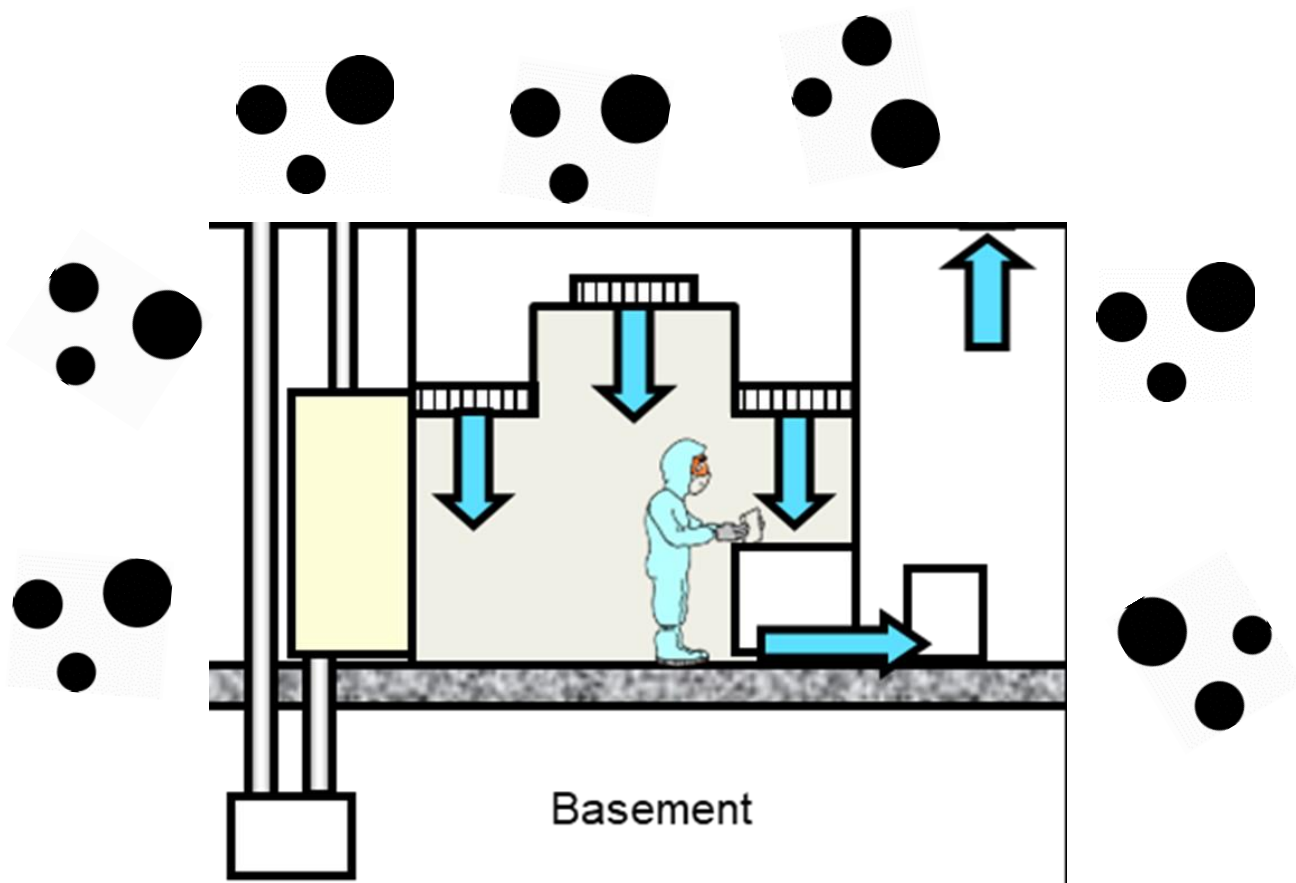
- air pollutant levels
- room pressure, temperature & humidity
- personnel access

to meet the critical environmental conditions that are demanded.

it's a

CONTROLLED ENVIRONMENT

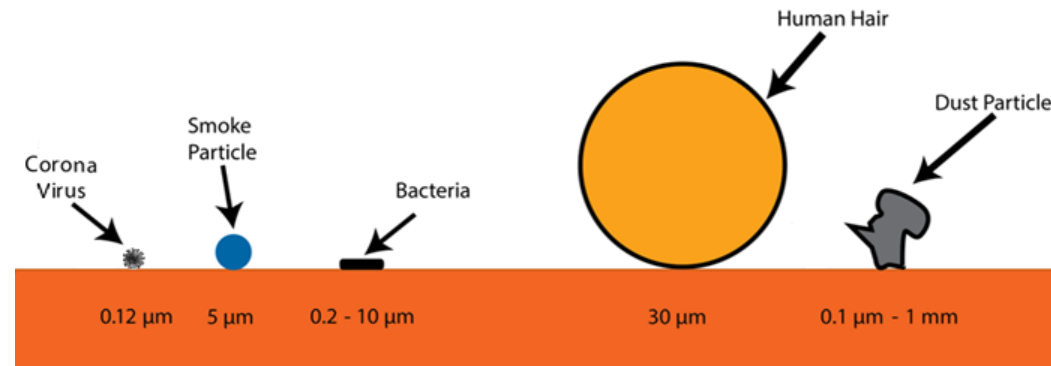
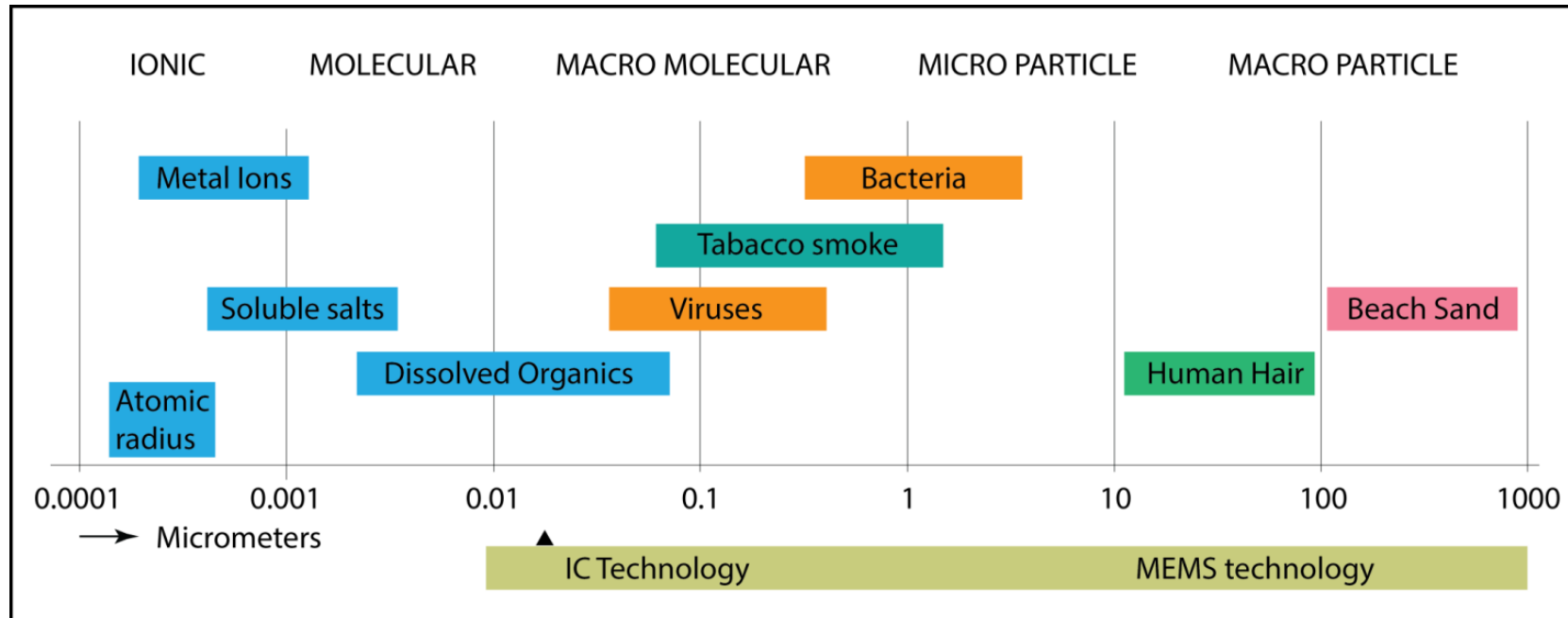
Contamination classification



Two classification systems are in use:

1. Federal Standard 209E
2. ISO 14644-1

Contamination size



Cleanroom classification systems

2 cleanroom classification systems are in use to indicate the level of cleanliness. They show the maximum allowed number of particles of a certain size in a certain amount of air for each level.

US FED STD 209E Cleanroom Standards						ISO 14644-1
Class	Maximum particles/ft ³					ISO equivalent
	≥ 0.1 μm	≥ 0.2 μm	≥ 0.3 μm	≥ 0.5 μm	≥ 5 μm	
1	35	7.5	3	1	0.007	ISO 3
10	350	75	30	10	0.07	ISO 4
100	3,500	750	300	100	0.7	ISO 5
1,000	35,000	7,500	3,000	1,000	7	ISO 6
10,000	350,000	75,000	30,000	10,000	70	ISO 7
100,000	3.5 x 10 ⁶	750,000	300,000	100,000	700	ISO 8

MEMS lab

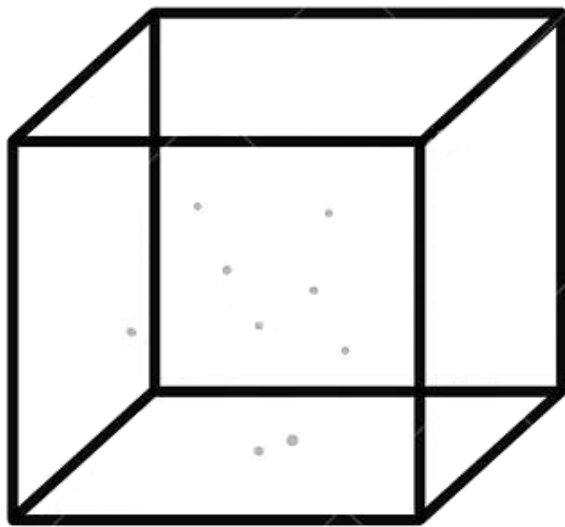


The numbers 1, 10, 100, 10000 denote the number of particles of size 0.5 μm or larger that are permitted per cubic foot of air.

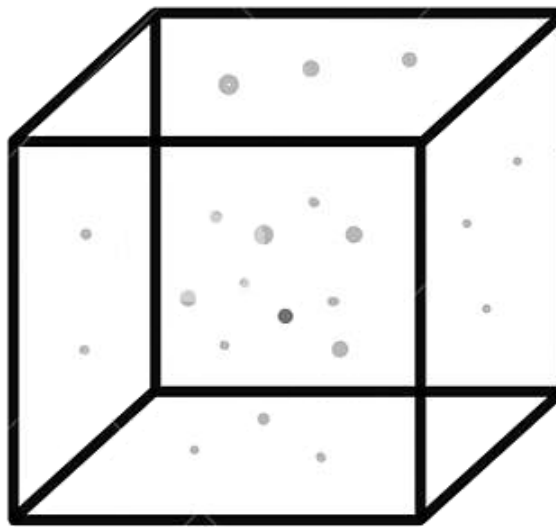
Class comparison

Number of Particles per Cubic Meter

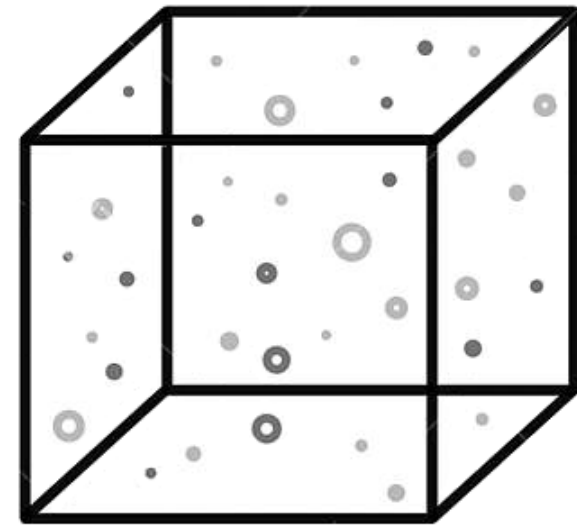
The level of cleanliness in the room by ISO 14644-1



Class ISO 5

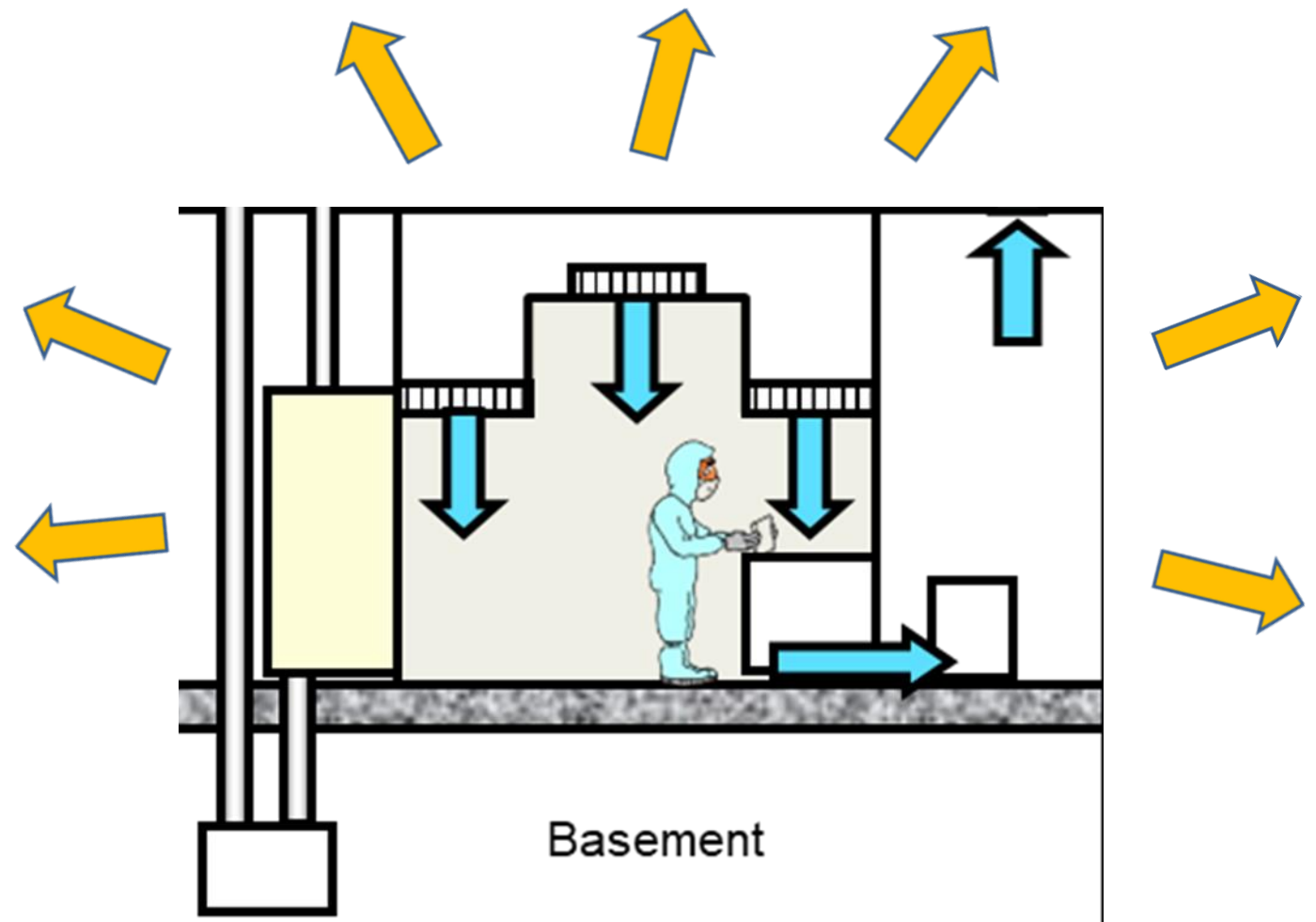
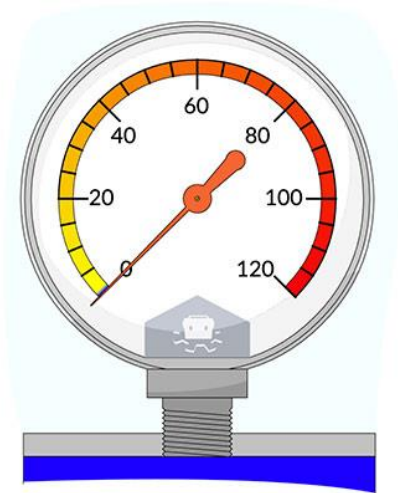


Class ISO 6



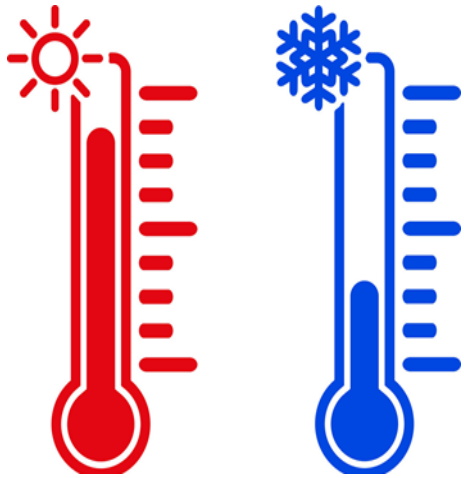
Class ISO 7

Pressurization



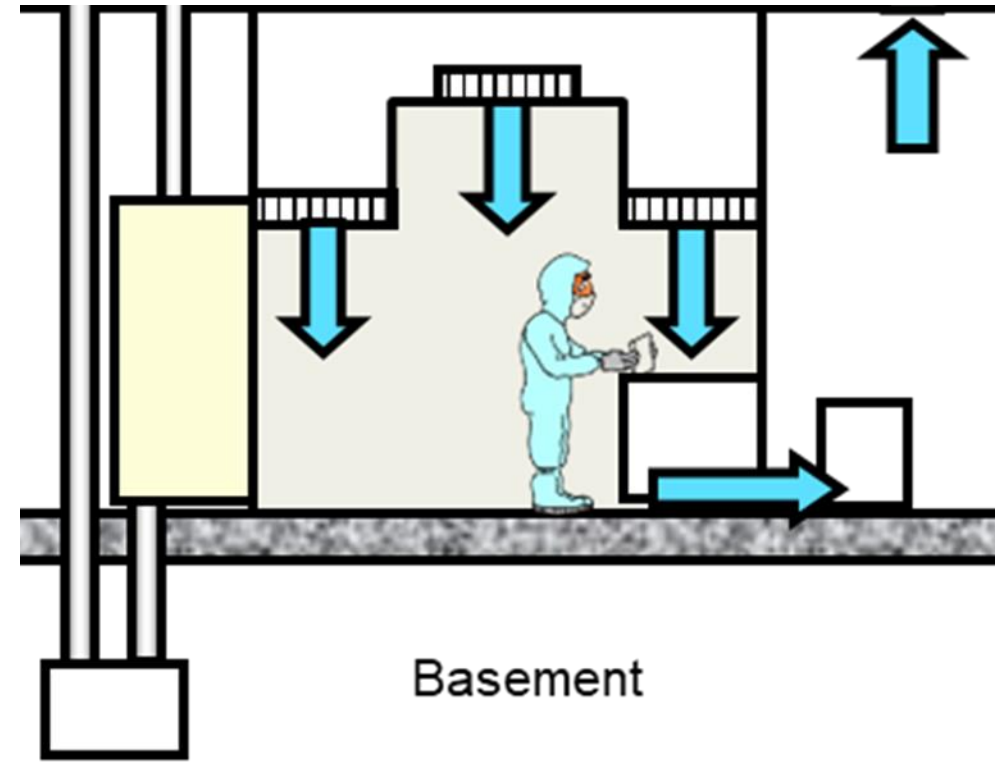
Pressurization is achieved by air pumps/fans, pumping air into the cleanroom through a HEPA filter to prevent dust from getting in. If anyone opens a door, air goes out instead of coming in from the outside.

Temperature and humidity

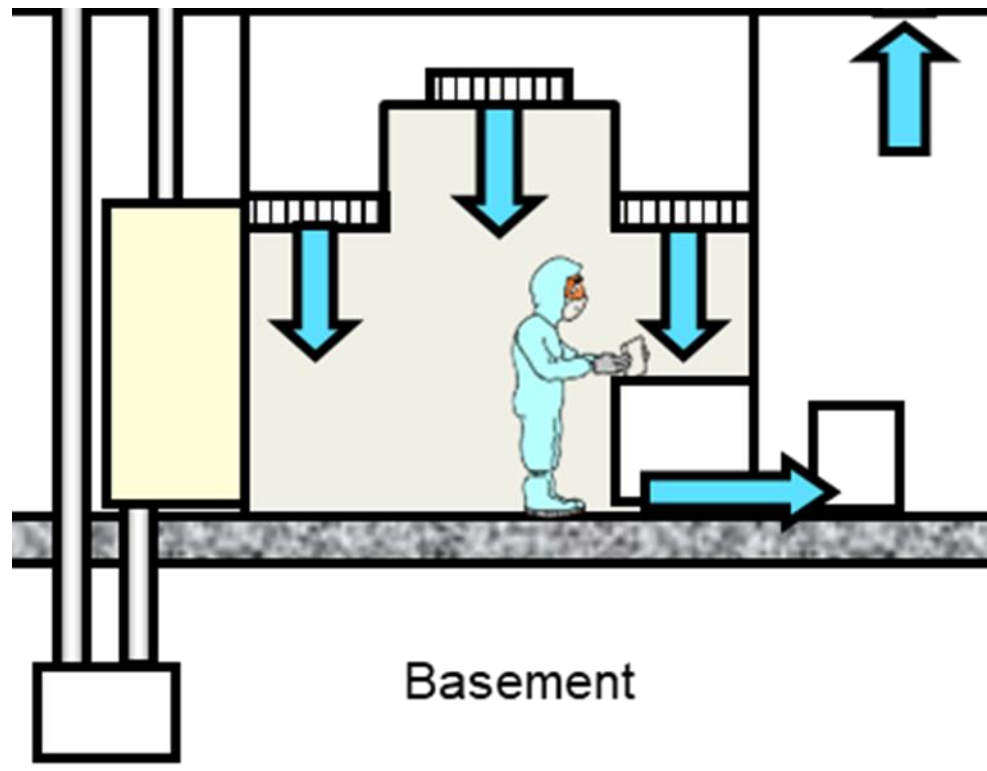
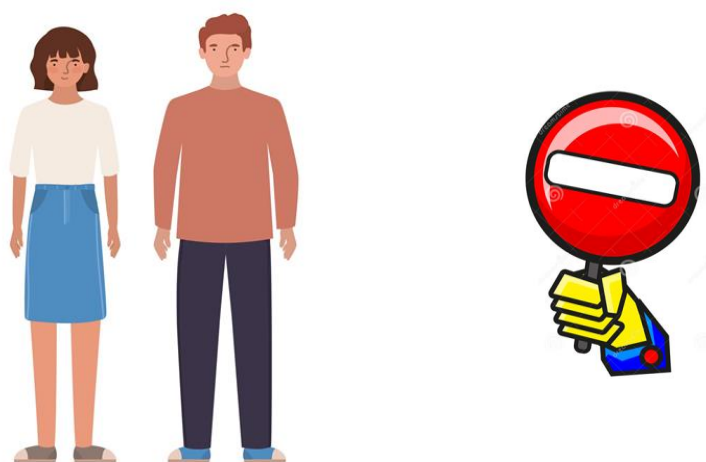


$T=21^{\circ}\text{C}+/-2$

RH at 30-55%



Personal access



EKL infrastructure

Total laboratory space: 1570m²

Cleanroom ISO 5 (38%)

(Managed by EKL.)

Cleanroom ISO 7 (26%)

(Main user: PVMD group. Lab manager: [Guangtao Yang](#).)

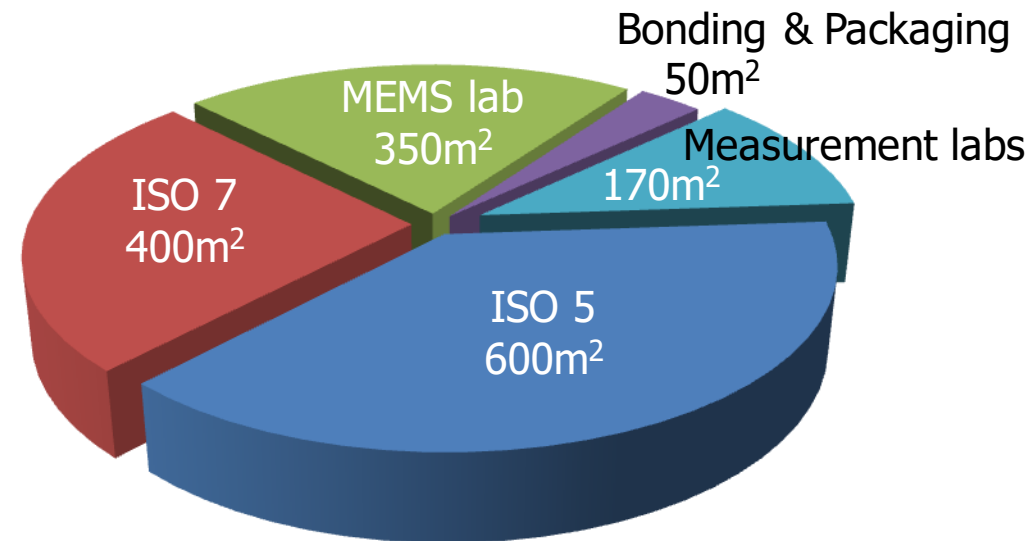
MEMS laboratory ISO 8 (22%)

(Partly managed by an external company.)

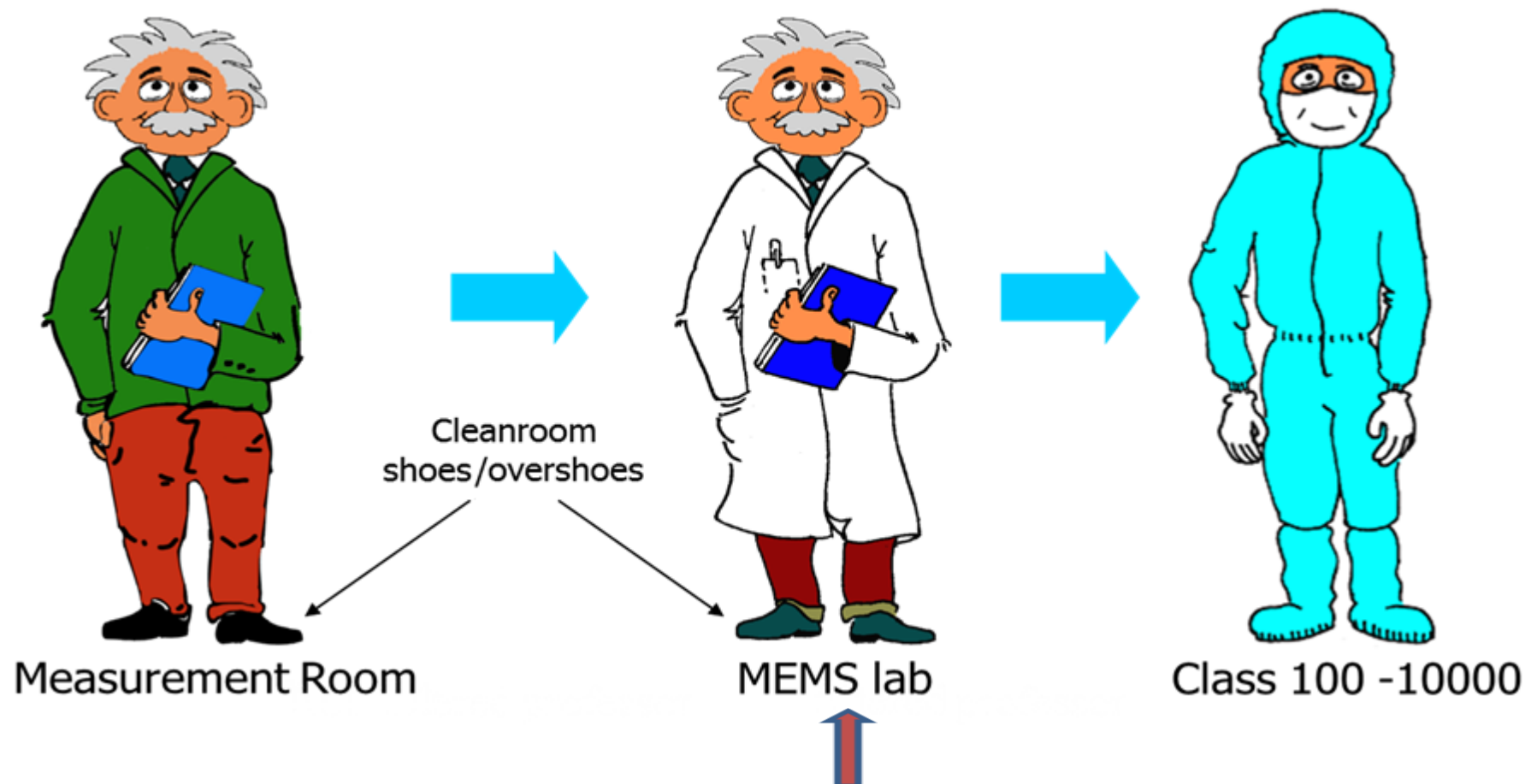
Electrical characterization laboratories (11%)

(Managed by EWI/ECTM. For access contact: [Filip Simjanoski](#).)

Bonding & Packaging laboratory (3%)



EKL lab clothing policy



CRYO lab exception: no lab coat needed, but overshoes required
(at least in the commonly used area with the overstep bench)

Equipment policy

- Access to the EKL labs **does not permit** the use of any equipment. Only certified persons are allowed to use the equipment in the labs unsupervised.
- Certification can only be obtained after an **Equipment Module training (MT)**
- A **Module Training** is equipment specific, and only includes instructions on how to use a system.
- **Users cannot be trained on a tool by staff members or users that are not authorized to train on that tool.**



Ethics

The success of each Lab User relies on:

- Understanding
- Shared responsibility among all users
- Participation
- Self discipline



EKL's rule violation protocol

- First time violation of rules:
 - Verbal warning.
- Second or moderate violation:
 - Talk with the Safety Officer
Re-training may be required.
- Third time or severe violation:
 - Lab access will be withdrawn.
 - Access will not be renewed, until an acceptable resolution is reached.
 - Re-training will be required.



